Intrinsic robust multiscale filtering

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Abstract

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1 Introduction

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Suppose that observed data Y is composed of a signal $X = (X_t)$ and a noise ε_t:

$$Y_t = X_t + \varepsilon_t .$$

2.1 Local contrast fit

Fix a contrast function $\rho(\cdot)$, a kernel function $K(\cdot)$, and a bandwidth h. For every t, compute

$$\widetilde{S}_t = \operatorname*{arginf}_x \sum_u \rho(Y_u - x) K_h(t - u) ,$$

where $K_h(y) = K(y/h)$.

For $\rho(y) = y^2/2$, the value $\widetilde{S}_t$ is the local mean of Y_u for $u \in [t-h, t+h]$:

$$\widetilde{S}_t = \frac{1}{N_h(t)} \sum_u Y_u \, K_h(t-u) , \qquad N_h(t) \stackrel{\mathrm{def}}{=} \sum_u K_h(t-u) ,$$

The choice $\rho(y) = |y|$ leads to the local median.

This step results in the first IMF $\widetilde{S}_t^{(1)} = \widetilde{S}_t$ and the residuals $Y_t^{(2)} = Y_t - \widetilde{S}_t$.

2.2 Iterative procedure

With $Y_t^{(1)} \stackrel{\mathrm{def}}{=} Y_t$, at every step $k \ge 1$, fix h_k and compute

$$\widetilde{S}_t^{(k)} = \operatorname*{arginf}_x \sum_u \rho(Y_u^{(k)} - x) K_{h_k}\bigl(t - u\bigr) , \quad t \in [0,1] .$$

(Here $Y_t^{(k)} = Y_{1+t}^{(k)}$ for all t.) Compute the new residuals

$$Y_t^{(k+1)} = Y_t^{(k)} - \widetilde{S}_t^{(k)} .$$

At every step k, decrease h_k by a fixed factor a (e.g. $a = \sqrt{2}$):

$$h_{k+1} = h_k / a .$$

Continue until $k \le k_{\max}$. Alternatively, fix the smallest bandwidth $h_{\min}$ and stop if $h_{k+1} < h_{\min}$.

2.3 Parameter choice and output

Sequence of bandwidths h_k start with $h_1 \asymp 1$ (very big, e.g. 0.1). The starting bandwidth should ensure that each window $[t - h_1, t + h_1]$ contains about half of all observations. This ensures that for the first IMF, almost all noise is smoothed out, and only gross features remain.

Kernel We apply the Epanechnikov kernel:

$$K(u) = \frac{3}{4}(1 - |u|)_+^2 .$$

Bandwidth factor The bandwidth factor a can be any value between one and two. Our default choice is $a = \sqrt{2}$ or $a = 2$.

Output The procedure delivers a sequence of IMF's

$$\widetilde{S}^{(1)}, \ \widetilde{S}^{(2)}, \ldots, \ \widetilde{S}^{(k_{\max})} .$$

This collection of the IMFs $(\widetilde{S}^{(k)})$ can be used for classification/identification.

2.4 True IMFs

The computed IMFs can be viewed as empirical/noisy versions of their population counterparts (true IMFs) defined with $X_t^{(1)} = X_t$ and the same sequence h_k by

$$S_t^{(k)} = \operatorname*{arginf}_x \sum_u \mathbb{E}\rho(Y_u^{(k)} - x) K_{h_k}(t - u) .$$

The question under study is the quality of recovery

$$\|\widetilde{S}^{(k)} - S^{(k)}\|^2 = \sum_{t=1}^{n} |\widetilde{S}_t^{(k)} - S_t^{(k)}|^2$$

or

$$\sup_t \bigl| \widetilde{S}_t^{(k)} - S_t^{(k)} \bigr| .$$

2.5 Theoretical study for local linear smoothing

Fix $\rho(u) = u^2/2$ corresponding to local mean procedure.

2.5.1 Linear representation of the IMFs

Define $W^{(k)} \colon I\!R^n \to I\!R^n$ as a linear mapping with the rows

$$W^{(k)} = \Bigl(\frac{1}{N_k(t)} K_{h_k}(t-u) \Bigr)_{t,u \in [0,1]} \tag{2.1}$$

with

$$N_k(t) \stackrel{\text{def}}{=} \sum_u K_{h_k}(t-u) \,.$$

Also, define

$$\mathcal{A}^{(k)} \stackrel{\text{def}}{=} (I\!I - W^{(k)}) \circ \ldots \circ (I\!I - W^{(1)})$$

$$\mathcal{W}^{(k)} \stackrel{\text{def}}{=} W^{(k)} \mathcal{A}^{(k-1)} \,.$$

Lemma 2.1. *Each* $\widetilde{S}^{(k)} \in I\!R^n$ *is a linear transform of the residuals* $Y^{(k)} \stackrel{\text{def}}{=} \mathcal{A}^{(k-1)} Y$ *:*

$$\widetilde{S}^{(k)} = W^{(k)} Y^{(k)} = W^{(k)} \mathcal{A}^{(k-1)} Y = \mathcal{W} Y \,, \qquad . \tag{2.2}$$

Proof. The identities $\widetilde{S}^{(k)} = W^{(k)} Y^{(k)}$ and $Y^{(k+1)} = Y^{(k)} - \widetilde{S}^{(k)}$ imply

$$Y^{(k+1)} = Y^{(k)} - \widetilde{S}^{(k)} = (I\!I - W^{(k)}) Y^{(k)} \,.$$

By induction, for $k \geq 1$,

$$Y^{(k+1)} = \prod_{j=1}^{k} (I\!I - W^{(j)}) Y = \mathcal{A}^{(k)} Y \,.$$

Similarly,

$$\widetilde{S}^{(k+1)} = W^{(k+1)} \prod_{j=1}^{k} (I\!I - W^{(j)}) Y = W^{(k+1)} \mathcal{A}^{(k)} Y = \mathcal{W} Y \,,$$

and the assertion follows. □

We can treat $(\mathcal{W}^{(k)})$'s as a family of multiscale operators in $I\!R^n$ mapping the data Y to a family of noisy IMFs $(\widetilde{S}^{(k)})$.

Lemma 2.2. *For every* $k \geq 1$, *the eigenvalues* $\lambda_j(W^{(k)})$ *of the operator* $W^{(k)}$ *satisfy*

$$\lambda_j(W^{(k)}) \in (0,1) \,, \qquad \| W^{(k)} \| = \lambda_{\max}(W^{(k)}) = 1 \,.$$

Furthermore, $\mathcal{A}^{(k)}$ *from* (2.2) *fulfills* $\| \mathcal{A}^{(k)} \| \leq 1$ *.*

Proof. Each matrix $W^{(k)}$ is stochastic is the sense that the elements of each row of $W^{(k)}$ are non-negative and sum to one. The result follows by Gershgorin theorem. □

2.5.2 Linear decomposition of the IMFs

Let $X_t = \mathbb{E} Y_t$, $X^{(1)} = X$.

Lemma 2.3. *For every $k \geq 1$, the true IMF $S^{(k)}$ can be represented as a linear transform of X:*

$$S^{(k)} = W^{(k)} \mathcal{A}^{(k-1)} X = W^{(k)} \circ (\mathbb{I} - W^{(k-1)}) \circ \ldots \circ (\mathbb{I} - W^{(1)}) X . \tag{2.3}$$

Let $\boldsymbol{\varepsilon} \stackrel{\text{def}}{=} X - S$ and $\mathbb{E}\boldsymbol{\varepsilon} = 0$. Then for $k \geq 1$,

$$\widetilde{S}^{(k)} - S^{(k)} = W^{(k)} \mathcal{A}^{(k-1)} \boldsymbol{\varepsilon} = W^{(k)} \boldsymbol{\varepsilon}^{(k-1)} ,$$

where $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^{(0)} \stackrel{\text{def}}{=} Y - X$ and

$$\begin{aligned} \boldsymbol{\varepsilon}^{(k)} &\stackrel{\text{def}}{=} (\mathbb{I} - W^{(k)}) \boldsymbol{\varepsilon}^{(k-1)} \\ &= (\mathbb{I} - W^{(k)}) \circ (\mathbb{I} - W^{(k-1)}) \circ \ldots \circ (\mathbb{I} - W^{(1)}) \boldsymbol{\varepsilon} = \mathcal{A}^{(k)} \boldsymbol{\varepsilon} . \end{aligned} \tag{2.4}$$

Proof. We can use

$$S^{(k)} = W^{(k)} X^{(k)} , \qquad X^{(k+1)} = X^{(k)} - S^{(k)} .$$

Statement (2.3) follows similarly to (2.2). □

2.5.3 Noise propagation

Let $\boldsymbol{\varepsilon}^{(k)} = \mathcal{A}^{(k)} \boldsymbol{\varepsilon}$ be given by (2.4). Obviously, $\mathbb{E}\boldsymbol{\varepsilon}^{(k)} = 0$. Define $\Sigma^{(k)} = \operatorname{Var}(\boldsymbol{\varepsilon}^{(k)})$.

Lemma 2.4. *It holds for $k \geq 1$*

$$\Sigma^{(k)} = \operatorname{Var}(\boldsymbol{\varepsilon}^{(k)}) = \mathcal{A}^{(k)} \operatorname{Var}(\boldsymbol{\varepsilon}) \mathcal{A}^{(k)\top} .$$

In particular, if $\operatorname{Var}(\varepsilon_t) = \sigma^2$ then $\Sigma^{(k)} = \mathcal{A}^{(k)} \mathcal{A}^{(k)\top}$. Furthermore, $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \Sigma)$ yields $\boldsymbol{\varepsilon}^{(k)} \sim \mathcal{N}(0, \Sigma^{(k)})$.

We can use

$$\mathbb{E} \| \widetilde{S}^{(k)} - S^{(k)} \|^2 = \mathbb{E} \| \boldsymbol{\varepsilon}^{(k)} \|^2 = \operatorname{tr} \Sigma^{(k)} .$$

Compute and plot $\operatorname{tr} \Sigma^{(k)} = \operatorname{tr}\bigl(\mathcal{A}^{(k)} \mathcal{A}^{(k)\top} \bigr)$ for $k = 1, \ldots, k_{\max}$.

Lemma 2.5. *It holds $\bigl(W^{(k)} W^{(k)\top} \bigr)_{tt} \asymp (n h_k)^{-1}$.*

2.5.4 Regular design

Assume a regular deterministic design $t = 1/n, 2/n, \ldots, 1$ and the vector $Y = (Y_t)$ is extended to $t \in [-0.5, 1.5]$; e.g. by $Y_t = Y_{1+t}$ for $t < 0$.

Lemma 2.6. *Assume a regular design with $N_k(t) \equiv N_k$ for $t \in [-0.5, 1.5]$. Then the operators $W^{(k)}$ from (2.1) and $\mathcal{A}^{(k)}$ from (2.2) mutually commute.*

Proof. Fix $k', k \geq 1$. Then for $t \in [0, 1]$

$$
\begin{aligned}
(W^{(k)} W^{(k')} Y)_t &= \sum_u \frac{K_{h_k}(t-u)}{N_k(t)} (W^{(k')} Y)_u \\
&= \sum_u \sum_s \frac{K_{h_k}(t-u)}{N_k(t)} \, \frac{K_{h_{k'}}(u-s)}{N_{k'}(u)} \, Y_s \\
&= \frac{1}{N_k \, N_{k'}} \sum_s \Bigl\{ \sum_u K_{h_k}(t-u) K_{h_{k'}}(u-s) \Bigr\} Y_s \, .
\end{aligned}
$$

The same representation applies to $(W^{(k')} W^{(k)} Y)_t$. The statement about $\mathcal{A}^{(k)}$ follows from (2.2). $\square$

3 Robust filtering

This section presents the robustified procedure based on local median filtering. More precisely, we apply a smoothed version of median filtering. Huber's proposal leads to

$$
\rho(x) \stackrel{\text{def}}{=} \frac{1}{2} x^2 \, \mathbb{I}(|x| \leq H) + (Hx - \frac{1}{2} H^2) \, \mathbb{I}(|x| > H) \, .
$$

The value H is a tuning meta-parameter. Our theoretical results require 3 derivatives of ρ. We consider the third-order spline with $\rho(0) = \rho'(0) = 0$, $\rho''(0) = 1$, and

$$
\rho'''(x) = -\operatorname{sign}(x) \, \mathbb{I}(H \leq |x| \leq 2H) \, .
$$

The function $\rho(x)$ is quadratic for $|x| \leq H$ with $\rho(x) = x^2/2$, then the second derivative $\rho(x)$ decreases from one to zero at $x = 2H$. As $\rho''(x) \geq 0$, this function is convex.

Alternatively, with $\rho_0(x) = |x|$ and $\phi_H(x) = \frac{1}{\sqrt{2\pi} H} \exp\Bigl(-\frac{x^2}{2H^2}\Bigr)$, one can use

$$
\rho_H(x) \stackrel{\text{def}}{=} \rho_0 * \phi_H(x) \, .
$$

This value can be computed analytically.

Lemma 3.1. *It holds*

$$\rho_H(x) = x\bigl\{2\Phi(x/H) - 1\bigr\} + \frac{2H}{\sqrt{2\pi}} \exp\bigl(-\frac{x^2}{2H^2}\bigr) . \tag{3.1}$$

Also, for $x \geq 0$,

$$\begin{aligned} \rho'_H(x) &= 2\Phi(x/H) - 1 , \\ \rho''_H(x) &= \frac{2}{H}\phi(x/H) = \frac{2}{H\sqrt{2\pi}} \exp\bigl(-\frac{x^2}{2H^2}\bigr) , \\ \rho'''_H(x) &= -\frac{2x}{H^3\sqrt{2\pi}} \exp\bigl(-\frac{x^2}{2H^2}\bigr) . \end{aligned}$$

Proof. We use

$$\rho_H(x) = \int_{-\infty}^{\infty} |t|\, \phi_H(x-t) dt = H \int_{-\infty}^{\infty} |t|\, \phi(x/H - t) dt = H \int_{-\infty}^{\infty} |t + x/H|\, \phi(t) dt .$$

For any $a \geq 0$,

$$\begin{aligned} \int_{-\infty}^{\infty} |t+a|\, \phi(t) dt &= \frac{1}{2} \int_{-\infty}^{\infty} \bigl(|t+a| + |t-a|\bigr)\, \phi(t) dt = a \int_{-a}^{a} \phi(t) dt + 2 \int_{a}^{\infty} t\, \phi(t) dt \\ &= a\bigl\{\Phi(a) - \Phi(-a)\bigr\} + \frac{2}{\sqrt{2\pi}} \int_{a}^{\infty} \mathrm{e}^{-t^2/2} d(t^2/2) = a\bigl\{\Phi(a) - \Phi(-a)\bigr\} + \frac{2}{\sqrt{2\pi}} \mathrm{e}^{-a^2/2} . \end{aligned}$$

This implies (3.1). The other identities follow by direct differentiation. $\square$

Lemma 3.2. *Check that* $\rho_H(x)$ *is*

- *infinitely differentiable;*
- *symmetric, convex, positive, and monotonously increasing with* $|x|$ *;*
- *it holds for all* x

$$\begin{aligned} |\rho'_H(x)| \leq\ 1 , \qquad 1 - |\rho'_H(x)|\ &\leq 2\bigl\{1 - \Phi(|x|/H)\bigr\} , \\ 0 \leq \rho''_H(x) \leq \frac{2}{H\sqrt{2\pi}}; \qquad |\rho'''_H(x)| &\leq \frac{2}{H^2\sqrt{2\pi}}; \end{aligned}$$

- *for* $|x| > 3H$ *,* $\bigl|\rho_H(x) - |x|\bigr|$ *is very small:*

$$\bigl|\rho_H(x) - |x|\bigr| \leq \frac{2H}{\sqrt{2\pi}} \exp\bigl(-\frac{x^2}{2H^2}\bigr) \leq 0.009H , \qquad |x| > 3H .$$

The procedure reads as follows:

- Set $Y^{(1)} \equiv Y$, $k = 1$;

- compute

$$
\begin{aligned}
\widetilde{S}_t^{(k)} &= \operatorname*{arginf}_x \sum_u \rho(Y_u^{(k)} - x) K_{h_k}(t-u)\,; \\
Y_t^{(k+1)} &= Y_t^{(k)} - \widetilde{S}_t^{(k)}\,;
\end{aligned}
\tag{3.2}
$$

- recompute $h_{k+1} = h_k / a$;

- increase k by 1; loop until $k > k_{\max}$ or $h_k < h_{\min}$;

Choice of H is important. This value should be sufficiently small, in particular, it should be smaller than the noise variance before contamination.

3.1 First-step decomposition

For a general ρ-function, one of the main questions is the definition of the true IMF. Define

$$
S_t \stackrel{\text{def}}{=} \operatorname*{arginf}_x \sum_u \mathbb{E}\rho(Y_u - x) K_h(t-u)\,.
$$

If $Y_t = X_t + \varepsilon_t$ for ε_t i.i.d. then

$$
\mathbb{E}\rho(Y_u - x) = \varphi(X_u - x)\,, \quad \varphi(z) \stackrel{\text{def}}{=} \mathbb{E}\rho(z + \varepsilon)\,,
$$

leading to

$$
S_t = \operatorname*{arginf}_x \sum_u \varphi(X_u - x) K_h(t-u)\,.
$$

Lemma 3.3. *ρ convex implies φ convex.*

Proof.

$$
\rho(\alpha\ z_1 + (1-\alpha) z_2) \le \alpha\rho + (1-\alpha)\rho(z_2)
$$

$$
\varphi(\alpha z_1 + (1 - alpha) z_2) =
$$

$$
= \mathbb{E}\rho(\alpha z_1 + (1-\alpha) z_2 + \varepsilon) =
$$

$$
= \rho(\alpha(z_1 + \varepsilon) + (1-\alpha)(z_2 + \varepsilon)) \le
$$

$$
\le E[\alpha\rho(z_1 + \varepsilon) + (1-\alpha)\rho(z_2 + \varepsilon)] =
$$

$$
= \alpha\varphi(z_1) + (1-\alpha)\varphi(z_2)
$$

Define

$$N_t(x) \stackrel{\text{def}}{=} \sum_u \mathbb{E}\rho''(Y_u - x)K_h(t-u) = \sum_u \varphi''(X_u - x)K_h(t-u) \tag{3.3}$$

and

$$N_t \stackrel{\text{def}}{=} N_t(X_t)\,.$$

Also, denote

$$\diamondsuit_{u,t} \stackrel{\text{def}}{=} \rho'(Y_u - X_t) - \mathbb{E}\rho'(Y_u - X_t) = \rho'(Y_u - X_t) - \varphi'(X_u - X_t)\,.$$

Obviously, $\mathbb{E}\diamondsuit_{u,t} \equiv 0$ *. The following expansion will be a key tool of analysis:*

$$\begin{aligned}\widetilde{S}_t - S_t \;&\approx\; \frac{1}{N_t}\sum_u \diamondsuit_{u,t}K_h(t-u) = \sum_u \diamondsuit_{u,t}\, w_{u,t}\,,\\ w_{u,t} &\stackrel{\text{def}}{=} \frac{K_h(t-u)}{N_t}\,.\end{aligned}$$

3.2 IMFs expansions

Set $Y^{(1)} \equiv Y$ *and* $X^{(1)} \equiv X$ *.*

Step $k \geq 1$ *: with* $\varepsilon_u^{(k)} \stackrel{\text{def}}{=} Y_u^{(k)} - X_u^{(k)}$

$$\begin{aligned}S_t^{(k)} &= \operatorname*{arginf}_x \sum_u \mathbb{E}\rho(Y_u^{(k)} - x)K_h(t-u)\\ &= \operatorname*{arginf}_x \sum_u \mathbb{E}\rho(X_u^{(k)} + \varepsilon_u^{(k)} - x)K_h(t-u)\\ &= \operatorname*{arginf}_x \sum_u \mathbb{E}\varphi_k(X_u^{(k)} - x, u)K_h(t-u)\,,\end{aligned}$$

where

$$\varphi_k(x,u) \stackrel{\text{def}}{=} \mathbb{E}\rho(x + \varepsilon_u^{(k)})\,.$$

Continue with $Y^{(k+1)} = Y^{(k)} - \widetilde{S}^{(k)}$ *from* (3.2) *and*

$$X^{(k+1)} = X^{(k)} - S^{(k)}\,.$$

4 Theoretical guarantees

This section describes the accuracy of the procedure. The starting point is an accuracy bounds for $\widetilde{S}_t - S_t$ *for a fixed* $t \in [0,1]$ *. We use the tools from perturbed optimization;*

see Section A. Represent

$$\widetilde{S}_t = \operatorname*{arginf}_x \sum_u \rho(Y_u - x) K_h(t-u)$$

$$S_t = \operatorname*{arginf}_x \sum_u \mathbb{E}\rho(Y_u - x)\, K_h(t-u) = \operatorname*{arginf}_x \sum_u \varphi(X_u - x) K_h(t-u) .$$

For t fixed, define

$$\begin{aligned} f(x) &\stackrel{\text{def}}{=} \sum_u \varphi(X_u - x) K_h(t-u), \\ f_\circ(x) &\stackrel{\text{def}}{=} \sum_u \rho(Y_u - x) K_h(t-u) . \end{aligned} \tag{4.1}$$

Obviously,

$$S_t = \operatorname*{arginf}_x f(x) , \qquad \widetilde{S}_t = \operatorname*{arginf}_x \{ f_\circ(x) \} .$$

Unfortunately, the stochastic component $\zeta(x)$ is not linear in x. We apply Theorem A.9 to f and $f_\circ$ from (4.1). The functions $f(x)$ and $f_\circ(x)$ are strongly convex in x. Denote $x^* = S_t = \operatorname{arginf}_x f(x)$, $x_\circ = \widetilde{S}_t = \operatorname{arginf}_x f_\circ(x)$, and define

$$\begin{aligned} \mathbb{F}_\circ &\stackrel{\text{def}}{=} f_\circ''(x^*) = \sum_u \rho''(Y_u - x^*) K_h(t-u) , \\ \mathbb{F} &\stackrel{\text{def}}{=} f''(x^*) = \sum_u \varphi''(X_u - x^*) K_h(t-u) . \end{aligned}$$

Note that $\mathbb{F} = N_t$; see (3.3). The key step of the proof is to show that for some small τ, it holds

$$\frac{1}{\mathbb{F}} \bigl| \mathbb{F}_\circ - \mathbb{F} \bigr| \le \tau .$$

One can use the representation

$$\mathbb{F}_\circ - \mathbb{F} = \sum_u \bigl\{ \rho''(Y_u - x^*) - \mathbb{E}\rho''(Y_u - x^*) \bigr\} K_h(t-u) .$$

This is a standard Bernstein-type deviation bound for independent sums under Lipschitz condition on each summand; see e.g. Boucheron et al. (2003).

define $D = \mathbb{F}^{1/2}$ and

$$\mathsf{b}_\circ \stackrel{\text{def}}{=} \frac{D}{\mathbb{F}_\circ} |f_\circ'(x^*)| = \frac{D}{\mathbb{F}_\circ} \sum_u \rho'(Y_u - x^*) K_h(t-u) .$$

As $f'(x^{*}) = 0$, *it also holds*

$$\mathsf{b}_{\circ} = \frac{D}{\mathbb{F}_{\circ}} \bigl\{ \rho'(Y_u - x^{*}) - \mathbb{E} \rho'(Y_u - x^{*}) \bigr\} K_h(t-u) \, .$$

Then by (A.8) *of Theorem A.9*

$$\bigl\| D^{-1} \mathbb{F}_{\circ} (x_{\circ} - x^{*} + \mathbb{F}_{\circ}^{-1} f_{\circ}'(x^{*})) \bigr\| \le \frac{3\tau_3}{4} \, \mathsf{b}_{\circ}^{2} \, . \tag{4.2}$$

This enables us to replace $\mathbb{F}_{\circ}$ *with* $\mathbb{F}$ *in the definition of* $\mathsf{b}_{\circ}$ *and to rewrite the bound* (4.2) *as*

$$\bigl\| D(x_{\circ} - x^{*} + \mathbb{F}_{\circ}^{-1} f_{\circ}'(x^{*})) \bigr\| \le \frac{3\tau_3}{4} \, \mathsf{b}_{\circ}^{2} \, , \qquad \mathsf{b}_{\circ} = \frac{1}{D} \sum_{u} \rho'(Y_u - x^{*}) K_h(t-u) \, .$$

A Smooth perturbed optimization

This section presents conditions and accurate bounds for smooth perturbed optimization. We consider minimization of a smooth convex function f and study the impact of a linear, quadratic, or smooth perturbation.

A.1 Gateaux smoothness and self-concordance

Below we assume the function $f(\boldsymbol{\upsilon})$, $\boldsymbol{\upsilon} \in \Upsilon \subseteq \mathbb{R}^p$ to be strongly convex with a positive definite Hessian $\mathbb{F}(\boldsymbol{\upsilon}) \stackrel{\text{def}}{=} \nabla^2 f(\boldsymbol{\upsilon})$. Also, assume $f(\boldsymbol{\upsilon})$ three or sometimes even four times Gateaux differentiable in $\boldsymbol{\upsilon} \in \Upsilon$. Local smoothness of f will be described by the relative error of the Taylor expansion of the third or fourth order. Define

$$\begin{aligned}
\delta_3(\boldsymbol{\upsilon}, \boldsymbol{u}) &= f(\boldsymbol{\upsilon} + \boldsymbol{u}) - f(\boldsymbol{\upsilon}) - \langle \nabla f(\boldsymbol{\upsilon}), \boldsymbol{u} \rangle - \frac{1}{2} \langle \nabla^2 f(\boldsymbol{\upsilon}), \boldsymbol{u}^{\otimes 2} \rangle, \\
\delta_3'(\boldsymbol{\upsilon}, \boldsymbol{u}) &= \langle \nabla f(\boldsymbol{\upsilon} + \boldsymbol{u}), \boldsymbol{u} \rangle - \langle \nabla f(\boldsymbol{\upsilon}), \boldsymbol{u} \rangle - \langle \nabla^2 f(\boldsymbol{\upsilon}), \boldsymbol{u}^{\otimes 2} \rangle,
\end{aligned}$$

and

$$\delta_4(\boldsymbol{\upsilon}, \boldsymbol{u}) = f(\boldsymbol{\upsilon} + \boldsymbol{u}) - f(\boldsymbol{\upsilon}) - \langle \nabla f(\boldsymbol{\upsilon}), \boldsymbol{u} \rangle - \frac{1}{2} \langle \nabla^2 f(\boldsymbol{\upsilon}), \boldsymbol{u}^{\otimes 2} \rangle - \frac{1}{6} \langle \nabla^3 f(\boldsymbol{\upsilon}), \boldsymbol{u}^{\otimes 3} \rangle .$$

Now, for each $\boldsymbol{\upsilon}$, suppose to be given a positive symmetric matrix $D(\boldsymbol{\upsilon})$ defining a local metric and a local vicinity around $\boldsymbol{\upsilon}$: for some radius $\mathtt{r}$,

$$\mathcal{U}_{\mathtt{r}}(\boldsymbol{\upsilon}) = \bigl\{ \boldsymbol{u} \in \mathbb{R}^p \colon \| D(\boldsymbol{\upsilon}) \boldsymbol{u} \| \le \mathtt{r} \bigr\}$$

Local smoothness properties of f at $\boldsymbol{\upsilon}$ are given via the quantities

$$\omega(\boldsymbol{\upsilon}) \stackrel{\text{def}}{=} \sup_{\boldsymbol{u} \colon \| D(\boldsymbol{\upsilon}) \boldsymbol{u} \| \le \mathtt{r}} \frac{2 |\delta_3(\boldsymbol{\upsilon}, \boldsymbol{u})|}{\| D(\boldsymbol{\upsilon}) \boldsymbol{u} \|^2}, \quad \omega'(\boldsymbol{\upsilon}) \stackrel{\text{def}}{=} \sup_{\boldsymbol{u} \colon \| D(\boldsymbol{\upsilon}) \boldsymbol{u} \| \le \mathtt{r}} \frac{|\delta_3'(\boldsymbol{\upsilon}, \boldsymbol{u})|}{\| D(\boldsymbol{\upsilon}) \boldsymbol{u} \|^2} . \tag{A.1}$$

The definition yields for any $\boldsymbol{u}$ with $\| D(\boldsymbol{\upsilon}) \boldsymbol{u} \| \le \mathtt{r}$

$$\bigl| \delta_3(\boldsymbol{\upsilon}, \boldsymbol{u}) \bigr| \le \frac{\omega(\boldsymbol{\upsilon})}{2} \| D(\boldsymbol{\upsilon}) \boldsymbol{u} \|^2 , \qquad \bigl| \delta_3'(\boldsymbol{\upsilon}, \boldsymbol{u}) \bigr| \le \omega'(\boldsymbol{\upsilon}) \| D(\boldsymbol{\upsilon}) \boldsymbol{u} \|^2 . \tag{A.2}$$

The approximation results can be improved under a third-order upper bound on the error of Taylor expansion.

$(\mathcal{T}_3)$ *For some τ_3*

$$\bigl| \delta_3(\boldsymbol{\upsilon}, \boldsymbol{u}) \bigr| \le \frac{\tau_3}{6} \| D(\boldsymbol{\upsilon}) \, \boldsymbol{u} \|^3 , \quad \bigl| \delta_3'(\boldsymbol{\upsilon}, \boldsymbol{u}) \bigr| \le \frac{\tau_3}{2} \| D(\boldsymbol{\upsilon}) \, \boldsymbol{u} \|^3 , \quad \boldsymbol{u} \in \mathcal{U}_{\mathtt{r}}(\boldsymbol{\upsilon}).$$

$(\mathcal{T}_4)$ *For some* τ_4

$$\bigl|\delta_4(\boldsymbol{v},\boldsymbol{u})\bigr| \le \frac{\tau_4}{24}\|D(\boldsymbol{v})\,\boldsymbol{u}\|^4, \qquad \boldsymbol{u}\in\mathcal{U}_{\mathtt{r}}(\boldsymbol{v}).$$

We also present a version of $(\mathcal{T}_3)$ *resp.* $(\mathcal{T}_4)$ *in terms of the third (resp. fourth) derivative of* f *.*

$(\mathcal{T}_3^*)$ $f(\boldsymbol{v})$ is three times differentiable and

$$\sup_{\boldsymbol{u}\colon \|D(\boldsymbol{v})\boldsymbol{u}\|\le\mathtt{r}} \sup_{\boldsymbol{z}\in\mathbb{R}^p} \frac{\bigl|\langle\nabla^3 f(\boldsymbol{v}+\boldsymbol{u}),\boldsymbol{z}^{\otimes 3}\rangle\bigr|}{\|D(\boldsymbol{v})\boldsymbol{z}\|^3} \le \tau_3 .$$

$(\mathcal{T}_4^*)$ $f(\boldsymbol{v})$ is four times differentiable and

$$\sup_{\boldsymbol{u}\colon \|D(\boldsymbol{v})\boldsymbol{u}\|\le\mathtt{r}} \sup_{\boldsymbol{z}\in\mathbb{R}^p} \frac{\bigl|\langle\nabla^4 f(\boldsymbol{v}+\boldsymbol{u}),\boldsymbol{z}^{\otimes 4}\rangle\bigr|}{\|D(\boldsymbol{v})\boldsymbol{z}\|^4} \le \tau_4 .$$

By Banach's characterization Banach (1938), $(\mathcal{T}_3^*)$ *implies*

$$\bigl|\langle\nabla^3 f(\boldsymbol{v}+\boldsymbol{u}),\boldsymbol{z}_1\otimes\boldsymbol{z}_2\otimes\boldsymbol{z}_3\rangle\bigr| \le \tau_3\|D(\boldsymbol{v})\boldsymbol{z}_1\|\,\|D(\boldsymbol{v})\boldsymbol{z}_2\|\,\|D(\boldsymbol{v})\boldsymbol{z}_3\|$$

for any $\boldsymbol{u}$ *with* $\|D(\boldsymbol{v})\boldsymbol{u}\|\le\mathtt{r}$ *and all* $\boldsymbol{z}_1,\boldsymbol{z}_2,\boldsymbol{z}_3\in\mathbb{R}^p$ *. Similarly under* $(\mathcal{T}_4^*)$

$$\begin{aligned}&\bigl|\langle\nabla^4 f(\boldsymbol{v}+\boldsymbol{u}),\boldsymbol{z}_1\otimes\boldsymbol{z}_2\otimes\boldsymbol{z}_3\otimes\boldsymbol{z}_4\rangle\bigr|\\ &\le \tau_4\|D(\boldsymbol{v})\boldsymbol{z}_1\|\,\|D(\boldsymbol{v})\boldsymbol{z}_2\|\,\|D(\boldsymbol{v})\boldsymbol{z}_3\|\,\|D(\boldsymbol{v})\boldsymbol{z}_4\|, \quad \boldsymbol{z}_1,\boldsymbol{z}_2,\boldsymbol{z}_3,\boldsymbol{z}_4\in\mathbb{R}^p .\end{aligned}$$

Lemma A.1. *Under* $(\mathcal{T}_3)$ *or* $(\mathcal{T}_3^*)$ *, it holds for* $\omega(\boldsymbol{v})$ *and* $\omega'(\boldsymbol{v})$ *from* (A.1)

$$\omega(\boldsymbol{v}) \le \frac{\tau_3\,\mathtt{r}}{3}, \qquad \omega'(\boldsymbol{v}) \le \frac{\tau_3\,\mathtt{r}}{2}.$$

Proof. For any $\boldsymbol{u}\in\mathcal{U}_{\mathtt{r}}(\boldsymbol{v})$ with $\|D(\boldsymbol{v})\boldsymbol{u}\|\le\mathtt{r}$

$$\bigl|\delta_3(\boldsymbol{v},\boldsymbol{u})\bigr| \le \frac{\tau_3}{6}\,\|D(\boldsymbol{v})\boldsymbol{u}\|^3 \le \frac{\tau_3\,\mathtt{r}}{6}\,\|D(\boldsymbol{v})\boldsymbol{u}\|^2,$$

and the bound for $\omega(\boldsymbol{v})$ follows. The proof for $\omega'(\boldsymbol{v})$ is similar. Under $(\mathcal{T}_3^*)$, apply the third order Taylor expansion to the univariate function $f(\boldsymbol{v}+t\boldsymbol{u})$ of t and $(\mathcal{T}_3^*)$ with $\boldsymbol{z}\equiv\boldsymbol{u}$. □

The values τ_3 *and* τ_4 *are usually very small. Some quantitative bounds are given later in this section under the assumption that the function* $f(\boldsymbol{v})$ *can be written in the form* $f(\boldsymbol{v})=nh(\boldsymbol{v})$ *for a fixed smooth function* $h(\boldsymbol{v})$ *with the Hessian* $\nabla^2 h(\boldsymbol{v})$ *. The factor* n *has meaning of the sample size; see Section* **??** *or Section* **??***.*

$(\mathcal{S}_3^*)$ $f(\boldsymbol{v}) = nh(\boldsymbol{v})$ for $h(\boldsymbol{v})$ three times differentiable and for a metric tensor $\boldsymbol{g}(\boldsymbol{v})$

$$\sup_{\boldsymbol{u}\colon \|\boldsymbol{g}(\boldsymbol{v})\boldsymbol{u}\| \le \mathtt{r}/\sqrt{n}} \frac{\bigl|\langle \nabla^3 h(\boldsymbol{v}+\boldsymbol{u}), \boldsymbol{u}^{\otimes 3}\rangle\bigr|}{\|\boldsymbol{g}(\boldsymbol{v})\boldsymbol{u}\|^3} \le \mathsf{c}_3 .$$

$(\mathcal{S}_4^*)$ the function $h(\cdot)$ satisfies $(\mathcal{S}_3^*)$ and

$$\sup_{\boldsymbol{u}\colon \|\boldsymbol{g}(\boldsymbol{v})\boldsymbol{u}\| \le \mathtt{r}/\sqrt{n}} \frac{\bigl|\langle \nabla^4 h(\boldsymbol{v}+\boldsymbol{u}), \boldsymbol{u}^{\otimes 4}\rangle\bigr|}{\|\boldsymbol{g}(\boldsymbol{v})\boldsymbol{u}\|^4} \le \mathsf{c}_4 .$$

$(\mathcal{S}_3^*)$ and $(\mathcal{S}_4^*)$ are local versions of the so-called self-concordance condition; see Nesterov and Nemirovskii (1994) and Ostrovskii and Bach (2021). In fact, they require that each univariate function $h(\boldsymbol{v}+t\boldsymbol{u})$ of $t \in \mathbb{R}$ is self-concordant with some universal constants c_3 and c_4. Under $(\mathcal{S}_3^*)$ and $(\mathcal{S}_4^*)$, with $D^2(\boldsymbol{v}) = n\,\boldsymbol{g}^2(\boldsymbol{v})$, the values $\delta_3(\boldsymbol{v},\boldsymbol{u})$, $\delta_4(\boldsymbol{v},\boldsymbol{u})$, and $\omega(\boldsymbol{v})$, $\omega'(\boldsymbol{v})$ can be well bounded.

Lemma A.2. *Suppose $(\mathcal{S}_3^*)$. Then $(\mathcal{T}_3)$ follows with $\tau_3 = \mathsf{c}_3 n^{-1/2}$. Moreover, for $\omega(\boldsymbol{v})$ and $\omega'(\boldsymbol{v})$ from (A.1), it holds*

$$\omega(\boldsymbol{v}) \le \frac{\mathsf{c}_3\,\mathtt{r}}{3n^{1/2}}, \qquad \omega'(\boldsymbol{v}) \le \frac{\mathsf{c}_3\,\mathtt{r}}{2n^{1/2}}. \tag{A.3}$$

Also $(\mathcal{T}_4)$ follows from $(\mathcal{S}_4^*)$ with $\tau_4 = \mathsf{c}_4 n^{-1}$.

Proof. For any $\boldsymbol{u} \in \mathcal{U}_{\mathtt{r}}(\boldsymbol{v})$, by third order Taylor expansion,

$$\begin{aligned}
|\delta_3(\boldsymbol{v},\boldsymbol{u})| &\le \sup_{t\in[0,1]} \frac{1}{6}\bigl|\langle \nabla^3 f(\boldsymbol{v}+t\boldsymbol{u}), \boldsymbol{u}^{\otimes 3}\rangle\bigr| = \frac{n}{6}\sup_{t\in[0,1]}\bigl|\langle \nabla^3 h(\boldsymbol{v}+t\boldsymbol{u}), \boldsymbol{u}^{\otimes 3}\rangle\bigr| \\
&\le \frac{n\,\mathsf{c}_3}{6}\,\|\boldsymbol{g}(\boldsymbol{v})\boldsymbol{u}\|^3 = \frac{n^{-1/2}\,\mathsf{c}_3}{6}\,\|D(\boldsymbol{v})\boldsymbol{u}\|^3 \le \frac{n^{-1/2}\,\mathsf{c}_3\,\mathtt{r}}{6}\,\|D(\boldsymbol{v})\boldsymbol{u}\|^2 .
\end{aligned}$$

This implies $(\mathcal{T}_3)$ as well as (A.3); see (A.2). The statement about $(\mathcal{T}_4)$ is similar. □

A.2 Perturbed optimization

This section provides an overview of some results from Spokoiny (2025). Let $f(\boldsymbol{v})$ be a smooth strongly convex function,

$$\boldsymbol{v}^* = \operatorname*{arginf}_{\boldsymbol{v}} f(\boldsymbol{v}),$$

and $\mathbb{F} = \nabla^2 f(\boldsymbol{v}^*)$. We study the question of how the point of minimum and the value of minimum of f change under some perturbation of f. More precisely, with $f^\circ(\boldsymbol{v})$

being a perturbed version of $f(\boldsymbol{\upsilon})$ *, we consider in parallel two optimization problems*

$$\boldsymbol{\upsilon}^{*} = \operatorname*{arginf}_{\boldsymbol{\upsilon}} f(\boldsymbol{\upsilon}), \quad \boldsymbol{\upsilon}^{\circ} = \operatorname*{arginf}_{\boldsymbol{\upsilon}} f^{\circ}(\boldsymbol{\upsilon}) .$$

The goal is to describe the discrepancies $\boldsymbol{\upsilon}^{\circ} - \boldsymbol{\upsilon}^{*}$ *and* $f^{\circ}(\boldsymbol{\upsilon}^{\circ}) - f^{\circ}(\boldsymbol{\upsilon}^{*})$ *induced by this perturbation. We also present the results for two specific cases: a linear and a quadratic perturbation.*

A.3 Smoothness conditions

Assume $f(\boldsymbol{\upsilon})$ *to be convex and three or sometimes even four times Gateaux differentiable in* $\boldsymbol{\upsilon} \in \Upsilon$ *. For a fixed* $\boldsymbol{\upsilon} \in \Upsilon$ *, we need such a condition uniformly in a local vicinity of a fixed point* $\boldsymbol{\upsilon} \in \Upsilon$ *. The notion of locality is given in terms of a metric tensor* $D(\boldsymbol{\upsilon}) \in \mathfrak{M}_p$ *, where* $\mathfrak{M}_p$ *is the set of symmetric positive* p *-matrices. In most of the results later on, one can use* $D(\boldsymbol{\upsilon}) = \mathbb{F}^{1/2}(\boldsymbol{\upsilon})$ *, where* $\mathbb{F}(\boldsymbol{\upsilon}) = \nabla^2 f(\boldsymbol{\upsilon})$ *. Introduce the following conditions.*

$(\boldsymbol{\mathcal{T}_3^*})$ $f(\boldsymbol{\upsilon}+\boldsymbol{u})$ is three times differentiable in $\boldsymbol{u}$ and with D possibly depending on $\boldsymbol{\upsilon}$

$$\sup_{\boldsymbol{u} \colon \|D\boldsymbol{u}\| \le \mathtt{r}} \sup_{\boldsymbol{z} \in \mathbb{R}^p} \frac{\bigl| \langle \nabla^3 f(\boldsymbol{\upsilon}+\boldsymbol{u}), \boldsymbol{z}^{\otimes 3} \rangle \bigr|}{\|D\boldsymbol{z}\|^3} \le \tau_3 .$$

$(\boldsymbol{\mathcal{T}_4^*})$ $f(\boldsymbol{\upsilon}+\boldsymbol{u})$ is four times differentiable in $\boldsymbol{u}$ and

$$\sup_{\boldsymbol{u} \colon \|D\boldsymbol{u}\| \le \mathtt{r}} \sup_{\boldsymbol{z} \in \mathbb{R}^p} \frac{\bigl| \langle \nabla^4 f(\boldsymbol{\upsilon}+\boldsymbol{u}), \boldsymbol{z}^{\otimes 4} \rangle \bigr|}{\|D\boldsymbol{z}\|^4} \le \tau_4 .$$

By Banach's characterization Banach (1938), $(\boldsymbol{\mathcal{T}_3^*})$ *implies*

$$\bigl| \langle \nabla^3 f(\boldsymbol{\upsilon}+\boldsymbol{u}), \boldsymbol{z}_1 \otimes \boldsymbol{z}_2 \otimes \boldsymbol{z}_3 \rangle \bigr| \le \tau_3 \|D\boldsymbol{z}_1\| \, \|D\boldsymbol{z}_2\| \, \|D\boldsymbol{z}_3\|$$

for any $\boldsymbol{u}$ *with* $\|D\boldsymbol{u}\| \le \mathtt{r}$ *and all* $\boldsymbol{z}_1, \boldsymbol{z}_2, \boldsymbol{z}_3 \in \mathbb{R}^p$ *.*

The value τ_3 *is usually very small. If the function* $f(\boldsymbol{\upsilon})$ *can be written in the form* $f(\boldsymbol{\upsilon}) = nh(\boldsymbol{\upsilon})$ *for a fixed smooth function* $h(\boldsymbol{\upsilon})$ *with the Hessian* $\nabla^2 h(\boldsymbol{\upsilon})$ *, and a factor* n *meaning of the sample size, then* $\tau_3 \asymp n^{-1/2}$ *.* $(\boldsymbol{\mathcal{T}_3^*})$ *is a local version of the so-called self-concordance condition; see Nesterov (1988) and Ostrovskii and Bach (2021).*

A.4 A linear perturbation

For the objective function f, *let another function* $f^{\circ}(\boldsymbol{\upsilon})$ *satisfy for some vector* $\boldsymbol{A}$

$$f^{\circ}(\boldsymbol{\upsilon}) - f^{\circ}(\boldsymbol{\upsilon}^{*}) = \bigl\langle \boldsymbol{\upsilon} - \boldsymbol{\upsilon}^{*}, \boldsymbol{A} \bigr\rangle + f(\boldsymbol{\upsilon}) - f(\boldsymbol{\upsilon}^{*}). \tag{A.4}$$

Define

$$\boldsymbol{\upsilon}^{\circ} \stackrel{\text{def}}{=} \operatorname*{arginf}_{\boldsymbol{\upsilon}} f^{\circ}(\boldsymbol{\upsilon}), \qquad f^{\circ}(\boldsymbol{\upsilon}^{\circ}) = \inf_{\boldsymbol{\upsilon}} f^{\circ}(\boldsymbol{\upsilon}). \tag{A.5}$$

The analysis aims to evaluate the quantities $\boldsymbol{\upsilon}^{\circ} - \boldsymbol{\upsilon}^{*}$ *and* $f^{\circ}(\boldsymbol{\upsilon}^{\circ}) - f^{\circ}(\boldsymbol{\upsilon}^{*})$.

A.4.1 Local concentration

Let f *satisfy* (A.1) *at* $\boldsymbol{\upsilon}^{*}$ *with* $D(\boldsymbol{\upsilon}^{*}) = D \le \varkappa\, \mathbb{F}^{1/2}$ *for some* $\varkappa > 0$. *The latter means that the matrix* $\mathbb{F} - \varkappa^{2} D^{2}$ *is positive definite. The next result describes the concentration properties of* $\boldsymbol{\upsilon}^{\circ}$ *from* (A.5) *in a local elliptic set*

$$\mathcal{A}(\mathtt{r}) \stackrel{\text{def}}{=} \{\boldsymbol{\upsilon} \colon \|\mathbb{F}^{1/2}(\boldsymbol{\upsilon} - \boldsymbol{\upsilon}^{*})\| \le \mathtt{r}\},$$

where $\mathtt{r}$ *is slightly larger than* $\|\mathbb{F}^{-1/2}\boldsymbol{A}\|$.

Theorem A.3. *Let* $f(\boldsymbol{\upsilon})$ *be a strongly convex function with* $f(\boldsymbol{\upsilon}^{*}) = \inf_{\boldsymbol{\upsilon}} f(\boldsymbol{\upsilon})$ *and* $\mathbb{F} = \nabla^{2} f(\boldsymbol{\upsilon}^{*})$. *Let, further,* $f^{\circ}(\boldsymbol{\upsilon})$ *and* $f(\boldsymbol{\upsilon})$ *be related by* (A.4) *with some vector* $\boldsymbol{A}$. *Fix* $\nu < 1$ *and* $\mathtt{r}$ *such that* $\|\mathbb{F}^{-1/2}\boldsymbol{A}\| \le \nu\,\mathtt{r}$. *Suppose now that* $f(\boldsymbol{\upsilon})$ *satisfy* (A.1) *for* $\boldsymbol{\upsilon} = \boldsymbol{\upsilon}^{*}$, $D(\boldsymbol{\upsilon}^{*}) = D \le \varkappa\, \mathbb{F}^{1/2}$ *with some* $\varkappa > 0$ *and* ω' *such that*

$$1 - \nu - \omega' \varkappa^{2} > 0.$$

Then for $\boldsymbol{\upsilon}^{\circ}$ *from* (A.5), *it holds*

$$\|\mathbb{F}^{1/2}(\boldsymbol{\upsilon}^{\circ} - \boldsymbol{\upsilon}^{*})\| \le \mathtt{r} \qquad \text{and} \qquad \|D(\boldsymbol{\upsilon}^{\circ} - \boldsymbol{\upsilon}^{*})\| \le \varkappa\,\mathtt{r}\,.$$

A.4.2 Second-order expansions

The result of Theorem A.3 allows to localize the point $\boldsymbol{\upsilon}^{\circ} = \operatorname{arginf}_{\boldsymbol{\upsilon}} f^{\circ}(\boldsymbol{\upsilon})$ *in the local vicinity* $\mathcal{A}(\mathtt{r})$ *of* $\boldsymbol{\upsilon}^{*}$. *The use of smoothness properties of* f° *or, equivalently, of* f, *in this vicinity helps to obtain rather sharp expansions for* $\boldsymbol{\upsilon}^{\circ} - \boldsymbol{\upsilon}^{*}$ *and for* $f^{\circ}(\boldsymbol{\upsilon}^{\circ}) - f^{\circ}(\boldsymbol{\upsilon}^{*})$.

Theorem A.4. *Assume the conditions of Theorem A.3 with* $\omega \le 1/3$. *Then*

$$\begin{aligned} -\frac{\omega}{1 - \varkappa^{2}\omega} \|D\,\mathbb{F}^{-1}\boldsymbol{A}\|^{2} &\le 2 f^{\circ}(\boldsymbol{\upsilon}^{\circ}) - 2 f^{\circ}(\boldsymbol{\upsilon}^{*}) + \|\mathbb{F}^{-1/2}\boldsymbol{A}\|^{2} \\ &\le \frac{\omega}{1 + \varkappa^{2}\omega} \|D\,\mathbb{F}^{-1}\boldsymbol{A}\|^{2}. \end{aligned}$$

Also

$$
\begin{aligned}
\|D(\boldsymbol{v}^{\circ} - \boldsymbol{v}^{*} + I\!\!F^{-1}\boldsymbol{A})\| &\leq \frac{2\sqrt{\omega}}{1 - \varkappa^2 \omega} \, \|D \, I\!\!F^{-1}\boldsymbol{A}\| \, , \\
\|D(\boldsymbol{v}^{\circ} - \boldsymbol{v}^{*})\| &\leq \frac{1 + 2\sqrt{\omega}}{1 - \varkappa^2 \omega} \, \|D \, I\!\!F^{-1}\boldsymbol{A}\| \, .
\end{aligned}
$$

A.4.3 Expansions under third order smoothness

The results of Theorem A.4 can be refined if f satisfies condition $(\mathcal{T}_3)$.

Theorem A.5. *Let $f(\boldsymbol{v})$ be a strongly convex function with $f(\boldsymbol{v}^{*}) = \inf_{\boldsymbol{v}} f(\boldsymbol{v})$ and $I\!\!F = \nabla^2 f(\boldsymbol{v}^{*})$. Let $f^{\circ}(\boldsymbol{v})$ fulfill (A.4) with some vector $\boldsymbol{A}$. Suppose that $f(\boldsymbol{v})$ follows $(\mathcal{T}_3)$ at $\boldsymbol{v}^{*}$ with D^2 , $\mathtt{r}$, and τ_3 such that*

$$
D^2 \leq \varkappa^2 \, I\!\!F, \quad \mathtt{r} \geq \frac{4\varkappa}{3} \, \|I\!\!F^{-1/2}\boldsymbol{A}\|, \quad \varkappa^3 \tau_3 \, \|I\!\!F^{-1/2}\boldsymbol{A}\| < \frac{1}{4} \, .
$$

Then $\boldsymbol{v}^{\circ} = \operatorname{arginf}_{\boldsymbol{v}} f^{\circ}(\boldsymbol{v})$ satisfies

$$
\|I\!\!F^{1/2}(\boldsymbol{v}^{\circ} - \boldsymbol{v}^{*})\| \leq \frac{4}{3} \|I\!\!F^{-1/2}\boldsymbol{A}\| \, , \qquad \|D(\boldsymbol{v}^{\circ} - \boldsymbol{v}^{*})\| \leq \frac{4\varkappa}{3} \, \|I\!\!F^{-1/2}\boldsymbol{A}\| \, .
$$

Moreover,

$$
\Bigl| 2f^{\circ}(\boldsymbol{v}^{\circ}) - 2f^{\circ}(\boldsymbol{v}^{*}) + \|I\!\!F^{-1/2}\boldsymbol{A}\|^2 \Bigr| \leq \frac{\tau_3}{2} \, \|D \, I\!\!F^{-1}\boldsymbol{A}\|^3 \, .
$$

A.4.4 Advanced approximation under locally uniform smoothness

The bounds of Theorem A.5 can be made more accurate if f follows $(\mathcal{T}_3^{*})$ *and* $(\mathcal{T}_4^{*})$ *and one can apply Taylor's expansion around any point close to $\boldsymbol{v}^{*}$. In particular, the improved results do not involve the value $\|I\!\!F^{-1/2}\boldsymbol{A}\|$ which can be large or even infinite in some situation; see Section A.5.*

Theorem A.6. *Let $f(\boldsymbol{v})$ be a strongly convex function with $f(\boldsymbol{v}^{*}) = \min_{\boldsymbol{v}} f(\boldsymbol{v})$ and $I\!\!F = \nabla^2 f(\boldsymbol{v}^{*})$. Assume $(\mathcal{T}_3^{*})$ at $\boldsymbol{v}^{*}$ with D^2 , $\mathtt{r}$, and τ_3 such that*

$$
D^2 \leq \varkappa^2 \, I\!\!F, \quad \mathtt{r} \geq \frac{3}{2} \|D \, I\!\!F^{-1}\boldsymbol{A}\| \, , \quad \varkappa^2 \tau_3 \|D \, I\!\!F^{-1}\boldsymbol{A}\| < \frac{4}{9} \, .
$$

Then

$$
\begin{aligned}
\|D(\boldsymbol{v}^{\circ} - \boldsymbol{v}^{*})\| &\leq \frac{3}{2} \|D \, I\!\!F^{-1}\boldsymbol{A}\| \, , \\
\|D^{-1} I\!\!F(\boldsymbol{v}^{\circ} - \boldsymbol{v}^{*} + I\!\!F^{-1}\boldsymbol{A})\| &\leq \frac{3\tau_3}{4} \|D \, I\!\!F^{-1}\boldsymbol{A}\|^2 \, .
\end{aligned}
\tag{A.6}
$$

Moreover,

$$\Big| 2 f^{\circ}(\boldsymbol{v}^{\circ}) - 2 f^{\circ}(\boldsymbol{v}^{*}) + \| \mathbb{F}^{-1/2} \boldsymbol{A} \|^{2} \Big| \leq \frac{\tau_3}{2} \, \| D \, \mathbb{F}^{-1} \boldsymbol{A} \|^{3} .$$

Remark A.1. The roles of f and f° can be exchanged. In particular, (A.6) applies with $\mathbb{F} = \mathbb{F}(\boldsymbol{v}^{\circ})$ provided that $(\mathcal{T}_3^*)$ is also fulfilled at $\boldsymbol{v}^{\circ}$.

A.5 Quadratic penalization

Here we discuss the case when $f^{\circ}(\boldsymbol{v}) - f(\boldsymbol{v})$ *is quadratic. The general case can be reduced to the situation with* $f^{\circ}(\boldsymbol{v}) = f(\boldsymbol{v}) - \| G \boldsymbol{v} \|^{2}/2$. *To make the dependence of* G *more explicit, denote* $f_G(\boldsymbol{v}) \stackrel{\text{def}}{=} f(\boldsymbol{v}) - \| G \boldsymbol{v} \|^{2}/2$,

$$\boldsymbol{v}^{*} = \operatorname*{arginf}_{\boldsymbol{v}} f(\boldsymbol{v}), \quad \boldsymbol{v}_G^{*} = \operatorname*{arginf}_{\boldsymbol{v}} f_G(\boldsymbol{v}) = \operatorname*{arginf}_{\boldsymbol{v}} \bigl\{ f(\boldsymbol{v}) - \| G \boldsymbol{v} \|^{2}/2 \bigr\}.$$

We study the bias $\boldsymbol{v}_G^{*} - \boldsymbol{v}^{*}$ *induced by this penalization. Now we turn to the general case with* f *satisfying* $(\mathcal{T}_3^*)$. *Define*

$$\mathbb{F}_G \stackrel{\text{def}}{=} \nabla^{2} f_G(\boldsymbol{v}^{*}), \quad \mathsf{b}_G \stackrel{\text{def}}{=} \| D \, \mathbb{F}_G^{-1} G^{2} \boldsymbol{v}^{*} \| .$$

Proposition A.7. *Let* $f_G(\boldsymbol{v}) = f(\boldsymbol{v}) - \| G \boldsymbol{v} \|^{2}/2$ *be convex and follow* $(\mathcal{T}_3^*)$ *with some* D^{2}, τ_3, *and* $\mathtt{r}$ *satisfying for* $\varkappa > 0$

$$D^{2} \leq \varkappa^{2} \, \mathbb{F}_G , \qquad \mathtt{r} \geq 3 \mathsf{b}_G / 2 , \qquad \varkappa^{2} \tau_3 \, \mathsf{b}_G < 4/9,$$

where $\mathsf{b}_G \stackrel{\text{def}}{=} \| D \, \mathbb{F}_G^{-1} G^{2} \boldsymbol{v}^{*} \|$. *Then*

$$\begin{aligned} \| D (\boldsymbol{v}_G^{*} - \boldsymbol{v}^{*}) \| &\leq \frac{3}{2} \, \mathsf{b}_G , \\ \bigl\| D^{-1} \mathbb{F}_G (\boldsymbol{v}_G^{*} - \boldsymbol{v}^{*} + \mathbb{F}_G^{-1} G^{2} \boldsymbol{v}^{*}) \bigr\| &\leq \frac{3 \tau_3}{4} \, \mathsf{b}_G^{2} , \\ \Big| 2 f_G(\boldsymbol{v}_G^{*}) - 2 f_G(\boldsymbol{v}^{*}) - \frac{1}{2} \| \mathbb{F}_G^{-1/2} G^{2} \boldsymbol{v}^{*} \|^{2} \Big| &\leq \frac{\tau_3}{2} \, \mathsf{b}_G^{3} . \end{aligned} \tag{A.7}$$

A.6 A smooth penalty

The case of a general smooth penalty $\operatorname{pen}_G(\boldsymbol{v})$ *can be studied similarly to the quadratic case. Denote* $f_G(\boldsymbol{v}) \stackrel{\text{def}}{=} f(\boldsymbol{v}) + \operatorname{pen}_G(\boldsymbol{v})$,

$$\boldsymbol{v}^{*} = \operatorname*{arginf}_{\boldsymbol{v}} f(\boldsymbol{v}), \quad \boldsymbol{v}_G^{*} = \operatorname*{arginf}_{\boldsymbol{v}} f_G(\boldsymbol{v}) = \operatorname*{arginf}_{\boldsymbol{v}} \bigl\{ f(\boldsymbol{v}) + \operatorname{pen}_G(\boldsymbol{v}) \bigr\}.$$

We study the bias $\boldsymbol{v}^*_G - \boldsymbol{v}^*$ *induced by this penalization. The statement of Theorem A.7 and its proof can be easily extended to this situation. Define*

$$\mathbb{F}_G \stackrel{\text{def}}{=} \nabla^2 f_G(\boldsymbol{v}^*), \quad \mathsf{b}_G \stackrel{\text{def}}{=} \| D\, \mathbb{F}_G^{-1} \boldsymbol{M}_G \| , \quad \boldsymbol{M}_G \stackrel{\text{def}}{=} \nabla \operatorname{pen}_G(\boldsymbol{v}^*) .$$

Theorem A.8. *Let* $f_G(\boldsymbol{v}) = f(\boldsymbol{v}) + \operatorname{pen}_G(\boldsymbol{v})$ *be strongly convex and follow* $(\mathcal{T}_3^*)$ *at* $\boldsymbol{v}^*$ *with some* D^2, τ_3, *and* $\mathtt{r}$ *satisfying for* $\varkappa > 0$

$$D^2 \le \varkappa^2 \, \mathbb{F}_G , \qquad \mathtt{r} \ge 3\mathsf{b}_G/2 , \qquad \varkappa^2 \tau_3 \, \mathsf{b}_G < 4/9,$$

Then

$$\| D(\boldsymbol{v}^*_G - \boldsymbol{v}^*) \| \le 3\mathsf{b}_G/2.$$

Moreover,

$$\begin{aligned} \bigl\| D^{-1} \mathbb{F}_G (\boldsymbol{v}^*_G - \boldsymbol{v}^* + \mathbb{F}_G^{-1} \boldsymbol{M}_G) \bigr\| &\le \frac{3\tau_3}{4} \, \mathsf{b}_G^2 , \\ \Bigl| 2 f_G(\boldsymbol{v}^*_G) - 2 f_G(\boldsymbol{v}^*) + \| \mathbb{F}_G^{-1/2} \boldsymbol{M}_G \|^2 \Bigr| &\le \frac{\tau_3}{2} \, \mathsf{b}_G^3 . \end{aligned}$$

Proof. Consider $f^\circ_G(\boldsymbol{v}) = f_G(\boldsymbol{v}) - \langle \nabla \operatorname{pen}_G(\boldsymbol{v}^*), \boldsymbol{v} \rangle$. Then f°_G is strongly convex and $\nabla f^\circ_G(\boldsymbol{v}^*) = 0$ yielding $\boldsymbol{v}^* = \operatorname{arginf}_{\boldsymbol{v}} f^\circ_G(\boldsymbol{v})$. Also, $f_G(\boldsymbol{v})$ is a linear perturbation of $f^\circ_G(\boldsymbol{v})$ with $\boldsymbol{A} = \boldsymbol{M}_G = \nabla \operatorname{pen}_G(\boldsymbol{v}^*)$. Now all the statements of Theorem A.7 apply to $\boldsymbol{v}^*_G$ with obvious changes. □

A.7 A generic perturbation

This section presents a generic perturbation lemma for (A.5) *without any assumption on the structure of the perturbation* $f^\circ(\boldsymbol{v}) - f(\boldsymbol{v})$. *Define*

$$\mathbb{F}_\circ \stackrel{\text{def}}{=} \nabla^2 f^\circ(\boldsymbol{v}^*), \quad \mathsf{b}_\circ \stackrel{\text{def}}{=} \| D\, \mathbb{F}_\circ^{-1} \nabla f^\circ(\boldsymbol{v}^*) \| .$$

Theorem A.9. *Let a function* $f^\circ(\boldsymbol{v})$ *be strongly convex and follow* $(\mathcal{T}_3^*)$ *at* $\boldsymbol{v}^*$ *with some* D^2, τ_3, *and* $\mathtt{r}$ *satisfying for* $\varkappa > 0$

$$D^2 \le \varkappa^2 \, \mathbb{F}_\circ , \qquad \mathtt{r} \ge 3\mathsf{b}_\circ/2 , \qquad \varkappa^2 \tau_3 \, \mathsf{b}_\circ < 4/9 .$$

Then

$$
\begin{aligned}
\|D(\boldsymbol{v}^{\circ} - \boldsymbol{v}^{*})\| &\leq \frac{3}{2}\,\mathsf{b}_{\circ}\,, \\
\bigl\|D^{-1}\mathbb{F}_{\circ}(\boldsymbol{v}^{\circ} - \boldsymbol{v}^{*} + \mathbb{F}_{\circ}^{-1}\nabla f^{\circ}(\boldsymbol{v}^{*}))\bigr\| &\leq \frac{3\tau_3}{4}\,\mathsf{b}_{\circ}^{2}\,, \\
\Bigl|2f^{\circ}(\boldsymbol{v}^{\circ}) - 2f^{\circ}(\boldsymbol{v}^{*}) + \|\mathbb{F}_{\circ}^{-1/2}\nabla f^{\circ}(\boldsymbol{v}^{*})\|^{2}\Bigr| &\leq \frac{\tau_3}{2}\,\mathsf{b}_{\circ}^{3}\,.
\end{aligned}
\tag{A.8}
$$

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